

TOI-1811b

R-2135

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TOI1811_260131 · created 2026-08-11T23:48:18+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2461072.0084 +/- 0.0059 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1614 +/- 0.0083
Transit depth (Rp/Rs)^2	2.61 +/- 0.27 [%]
Orbital Inclination (inc)	86.93 +/- 0.93
Airmass coefficient 1 (a1)	1.254 +/- 0.011
Airmass coefficient 2 (a2)	-0.22 +/- 0.1
Scatter in the residuals of the lightcurve fit is	0.87 %
Best Comparison Star	#2 - [620, 174]
Optimal Aperture	3.71
Optimal Annulus	10.0
Transit Duration (day)	0.087 +/- 0.014

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1614 $\pm$ 0.0083 vs 0.13272 (deviation 3.46 σ)
Duration fit vs published	0.087 d vs 0.0813 d (deviation 0.41 σ)
Mid-time O-C	-10.2 min
Depth SNR	19.4
Residual scatter	0.87 %

Light curve

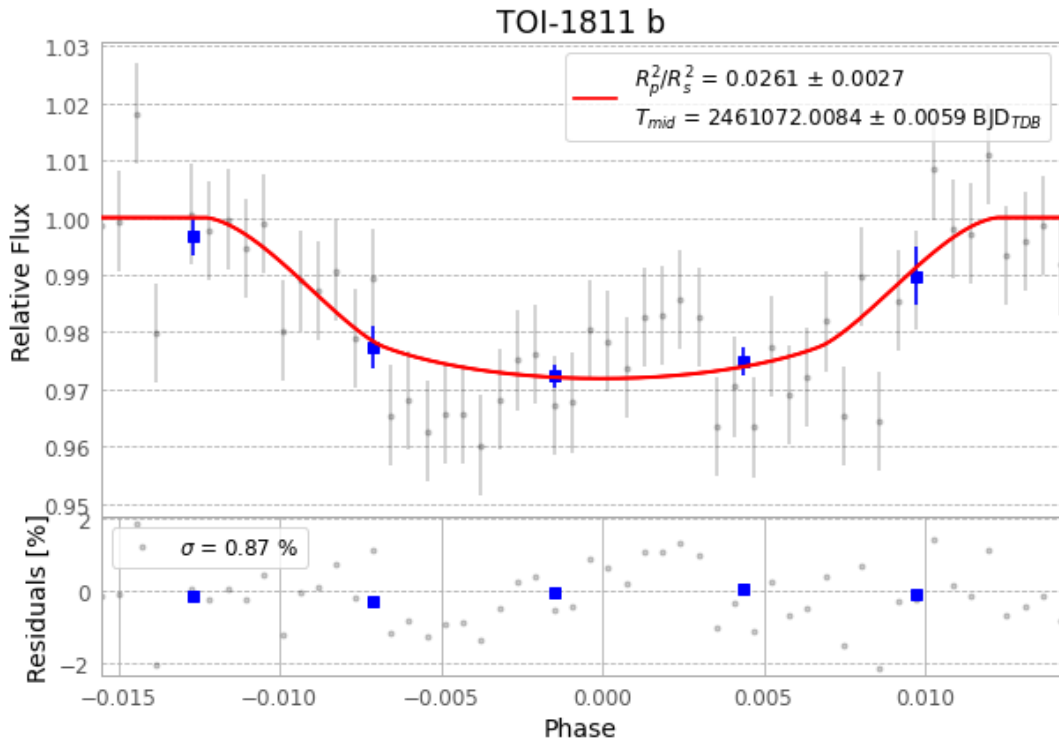


Figure 1 — FinalLightCurve_TOI-1811 b_2026-01-31.png (SHA-256 cd76f7cb08cfa67f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [342, 119], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 60e247ce44515697...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

54 FITS frames (35 MB), TOI1811260131104216.FITS → TOI1811260131132115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TOI1811-260131.log (5354bc180c94...), FinalLightCurve_TOI-1811 b_2026-01-31.png (cd76f7cb08cf...), FinalLightCurve_TOI-1811 b_2026-01-31.csv (15228085b4ae...)

Compute resources

Wall time 136.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-11 23:48Z.